

FinKont2: Hand-In Exercise #2

Answers (individually composed – but feel free to discuss in groups) must be handed in via Absalon no later than 23:59 CET on Monday March 9, 2026. There are 10 questions in the following each worth 10 points.

American options: The finite case

Consider first the finite probabilistic (so: no finance yet) setting from section 2.1-3 in Lambertson & Lapeyre (2008)

1.a] Prove their Proposition 2.2.1; making stopped Snell envelopes martingales.

1.b] Prove their Proposition 2.3.1; Doob decomposition of supermartingales.

Consider now the finance setting/application from Poulsen (2017), in particular the point 1.-2. argumentation/proof on the top of the second column on the first page.

1.c] Add a point 3. wherein you carefully argue that if $\pi_t^{\text{AMR}} = Z_t$, then no arbitrage is possible.

1.d] Extend the argumentation in points 1.-2. to the case where the interest rate is non-zero, thus fully proving equation (*). Why is that of considerable financial relevance?

1.e] (For linguistic ease, “discounting”/”discounted” has been omitted in the following three sentences.) In point 2. of the proof we construct an arbitrage by selling the American option and buying a self-financing replicating strategy for the martingale part of the Doob decomposition of the Snell envelope of the intrinsic value of the option. We know that the optimally stopped Snell envelope is itself a Q -martingale, and can thus be replicated by a self-financing strategy. Construct a counter-example showing that doing this does not guarantee us an arbitrage profit – i.e. the Doob decomposition part is necessary. Hint: Here is an example build from the standard binomial model.

American options: Longstaff & Schwartz

The following questions are closely related to

Poulsen (2025), “Secrets of Longstaff & Schwartz aged 23 3/4: Part I, the example”.

2.a Table 1 in Poulsen (2025): Explain and replicate the numbers (which you might want to do in reverse order). Remark: If you are cleverer than I were when I wrote the column (say, you are Niels Richard), you can actually find $\hat{\sigma}_{ML}$ explicitly in closed form – but this is not a MatStat exam!

2.b Table 2 in Poulsen (2025): Explain and replicate the numbers.

2.c Solve question 1 from the ‘Exercises for the reader’-section in Poulsen (2025).

2.d Solve question 2 from the ‘Exercises for the reader’-section in Poulsen (2025).

2.e Solve question 3 from the ‘Exercises for the reader’-section in Poulsen (2025).